1. (a)	Attempts gradient = $\frac{20--4}{-5-3}=(-3)$	M1
	Attempts equation of line $y-20="-3"(x+5)$ or $y+4="-3"(x-3)$	dM1
	$y=-3x+5$	A1
		(3)
(b)	Gradient $\frac{1}{3}$ or midpoint $\left(-1,8\right)$	B1ft
	$y-8=\frac{1}{3}(x+1)$	M1
	$x-3y+25=0$	A1
		(3)
		(6 marks)

2. (i)	$\dfrac{3y^3\left(2x^4\right)^3}{4x^2y^4}=\dfrac{6x^{10}}{y}$	B1, B1, B1 **(3)**
(ii)	$\dfrac{16}{\sqrt{3}+1}=a\sqrt{27}+4$	
	States or uses $\sqrt{27}=3\sqrt{3}$	B1
	Correct attempt at rationalising seen e.g. $\dfrac{16}{\sqrt{3}+1}\times\dfrac{\sqrt{3}-1}{\sqrt{3}-1}$	M1
	Correct attempt to make "a" the subject and make progress towards the form $p\sqrt{3}+q$ e.g. $8\sqrt{3}-8=3a\sqrt{3}+4\Rightarrow 3a\sqrt{3}=8\sqrt{3}-12\Rightarrow a=\dfrac{8\sqrt{3}}{3\sqrt{3}}-\dfrac{12}{3\sqrt{3}}$	M1
	$a=-\dfrac{4}{3}\sqrt{3}+\dfrac{8}{3}$	A1cso **(4)** **(7 marks)**

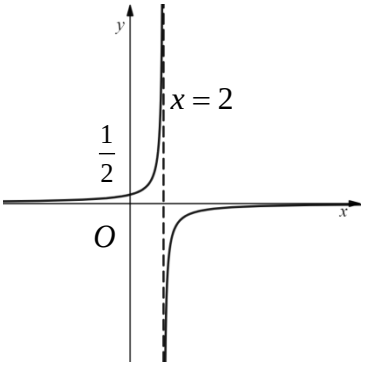
3. (a)	$\int \mathrm{f}(x)\,\mathrm{d}x = \int x^{\frac{3}{2}} + 10x^{\frac{1}{2}} + 25x^{-\frac{1}{2}}\,\mathrm{d}x = \frac{2}{5}x^{\frac{5}{2}} + \frac{20}{3}x^{\frac{3}{2}} + 50x^{\frac{1}{2}} + c$	M1 A1 A1 A1 **(4)**
(b) (i)	$\mathrm{f}'(x) = \frac{3}{2}x^{\frac{1}{2}} + 5x^{-\frac{1}{2}} - \frac{25}{2}x^{-\frac{3}{2}}$	M1 A1
	$\mathrm{f}'(x) = 0 \Rightarrow \frac{3}{2}x^{\frac{1}{2}} + 5x^{-\frac{1}{2}} - \frac{25}{2}x^{-\frac{3}{2}} = 0$ AND $\times x^{\frac{3}{2}}$	dM1
	$\frac{3}{2}x^{2} + 5x^{1} - \frac{25}{2} = 0 \Rightarrow 3x^{2} + 10x - 25 = 0$ *	A1*
(ii)	$\left(x =\right)\frac{5}{3}$ only	B1 **(5)** **(9 marks)**

4 (a)	States or implies that $\left[f(x) = \right] kx(x-4)$	M1
	Attempts to find k. E.g. $-4.8 = k \times 2 \times (2-4) \Rightarrow k = ...$	dM1
	$\left[f(x) = \right] 1.2x(x-4)$	A1
		(3)
(b)	States or implies that $\left[g(x) = \right] \lambda x(x-4)^2$	M1
	Attempts to find λ . E.g. $7.2 = \lambda \times 6 \times (6-4)^2 \Rightarrow \lambda = ...$	dM1
	$\left[g(x) = \right] 0.3x(x-4)^2$	A1
		(3)
(c)	Sets their $1.2x(x-4) = 0.3x(x-4)^2$	B1ft
	Valid attempt to solve $1.2\cancel{x}\cancel{(x-4)} = 0.3\cancel{x}(x-4)^{\cancel{2}} \Rightarrow x = 4 + \frac{1.2}{0.3}$	M1
	$x = 8$	A1
	$(8, 38.4)$	A1
		(4)
		(10 marks)

5 (a)	Attempts $r\theta = 5\times 1.2$ Perimeter =5 + 5 + 6 = 16 (km)	M1 A1 **(2)**
(b)	Attempts $\frac{1}{2}r^2\theta = \frac{1}{2}\times 5^2\times 1.2$	M1
	Area $AOP = \frac{1}{4}\times\left(\frac{1}{2}\times 5^2\times 1.2\right) = 3.75\,\text{km}^2$ *	M1, A1* **(3)**
(c)	Sets $\frac{1}{2}\times 5\times OP\times\sin 1.2 = 3.75 \Rightarrow OP = ...$	M1
	$OP = 1.6...$	A1
	$AP^2 = 5^2 + "1.6.."^2 - 2\times 5\times "1.6.."\times\cos 1.2$	M1
	$AP = 4.7\text{km}$ or 4700m	A1cso **(4)** **(9 marks)**

5(c) Alt	Let X be base of perpendicular to OA though P, then $\frac{1}{2}\times 5\times XP = 3.75 \Rightarrow XP = 1.5 \Rightarrow OX = \frac{"1.5"}{\tan 1.2}$ so $AX = 5 - \frac{"1.5"}{\tan 1.2} = ...$ $AX = 4.416...$ $AP^2 = "4.416..."^2 + "1.5"^2 = ...$ $AP = 4.7\text{km}$ or 4700m	M1 A1 M1 A1 **(4)** **(9 marks)**

6.(a)



1/*x* type shape

Fully correct

Correct equation of vertical asymptote and *y* intercept

M1
A1
B1

(3)

(b) (i) Sets $kx-4=\dfrac{1}{2-x}\Rightarrow(kx-4)(2-x)=1$ M1

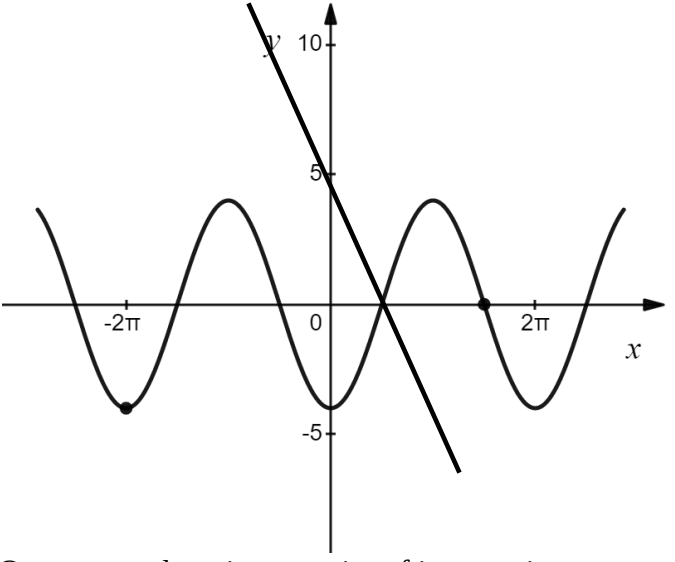
$kx^2+(-4-2k)x+9=0$ A1

Attempts use $b^2-4ac\ldots0\ \Rightarrow(-4-2k)^2-4k\times9\ldots0$ dM1

$\Rightarrow 4k^2-20k+16\ldots0\Rightarrow k^2-5k+4\ldots0$ * A1*

(ii) $(k-1)(k-4)\ldots0\Rightarrow k$,, 1, $k\ldots4$ M1, A1

(6)
(9 marks)

7 (a)(i)	$P = (-2\pi, -4)$	B1, B1
(ii)	$Q = \left(\frac{3\pi}{2}, 0\right)$	B1 **(3)**
(b) (i)	$k = 7$	B1
(ii)	$(2\pi, 3)$	M1 A1 **(3)**
(c)	 One root as there is one point of intersection	M1A1 **(2)** **(8 marks)**

8 (a)	States or implies that gradient of tangent is 24	B1
	Solves $f'(3)=24$ to find k. E.g $4\times3^2+k\times3+3=24\Rightarrow k=...$	M1
	$3k+39=24\Rightarrow k=\dfrac{24-39}{3}=-5$ *	A1*
		(3)
(b)	$f'(x)=4x^2-5x+3\Rightarrow f(x)=\dfrac{4}{3}x^3-\dfrac{5}{2}x^2+3x+c$	M1 A1
	Substitutes $x=3, y=-\dfrac{3}{24}+5$ into $y=f(x)$ to find "c"	dM1
	$f(x)=\dfrac{4}{3}x^3-\dfrac{5}{2}x^2+3x-\dfrac{141}{8}$	A1
		(4)
		(7 marks)

9 (a)	$x \ldots -5$	B1 **(1)**
(b)	$f(x) = (x+5)\left(3x^2 - 4x + 20\right) = 3x^3 + 11x^2 + 100$	M1
	$f'(x) = 9x^2 + 22x$	M1 A1cso **(3)**
(c)	Finds $f'(-4) = 9\times\left(-4\right)^2 + 22\times -4 = \left(56\right)$	M1
	Sets $f'(x) = "9x^2 + 22x" = "56"$	dM1
	$9x^2 + 22x - 56 = 0 \Rightarrow x = \frac{14}{9}, \left(-4\right)$	ddM1 A1cso **(4)**
(d)(i)	$\left(-1, 84\right)$	B1
(ii)	$\left(-4, 336\right)$	B1 **(2)** **(10 marks)**